THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 11: **Electricity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Why are household appliances connected in parallel? 1

(a) it saves power
(b) it reduces current
(c) it increases voltage
(d) Each gets the same voltage and works independently

2. Conventional current flows from 1

(a) only in the cell
(b) the negative to the positive terminal
(c) in either direction
(d) the positive terminal to the negative terminal outside the cell

3. Which material is used in the filament of an electric heater? 1

(a) Nichrome **(b)** Copper **(c)** Silver **(d)** Aluminium

4. The heat produced in a resistor is given by 1

(a) I R^2 t **(b)** V I / t **(c)** I^2 R t **(d)** I R / t

5. The charge on an electron is 1

(a) 9.1×10^{-31} C **(b)** 1.6×10^{-19} C **(c)** 6.25×10^{18} C **(d)** 1.6×10^{19} C

6. Which of the following has the lowest resistivity? 1

(a) Mercury **(b)** Silver **(c)** Nichrome **(d)** Iron

7. The SI unit of electric power is 1

(a) ampere **(b)** watt **(c)** joule **(d)** volt

8. The device that maintains a potential difference across a conductor is 1

(a) a switch **(b)** a resistor
(c) a cell or battery **(d)** an ammeter

9. The resistance of a wire is R. If its length is doubled and its area is halved, the new resistance is 1

(a) 2R **(b)** R **(c)** 4R **(d)** R/2

10. 1 kilowatt hour equals 1

(a) 3.6×10^3 J **(b)** 36×10^6 J **(c)** 1000 J **(d)** 3.6×10^6 J

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SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do we use a fuse in electrical circuits? 2

12. Why is an ammeter connected in series and a voltmeter in parallel? 2

13. What is the effect on resistance when the length of a wire is doubled? 2

14. What is the commercial unit of electric energy? Give its value in joules. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the resistance of a conductor increases with its length and decreases with its area. 3

16. An electric iron of 1100 W is operated at 220 V. Find the current and the resistance. 3

17. Explain why the series arrangement is not used for household circuits. 3

OR

The resistance of a wire of length 2 m and area 0.5 mm^2 is 4 ohm. Find its resistivity. 3

18. Calculate the heat produced by a 10 ohm resistor carrying 2 A current for 30 s. 3

OR

What is the resistance of a bulb rated 100 W, 220 V? Find the current. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Define resistivity. State the factors on which it depends. A wire of length 1 m and area of cross-section 1 mm^2 has resistance 1.5 ohm. Find its resistivity. 5

OR

An electric heater of 2 kW is used for 2 hours daily for 30 days at Rs 6 per unit. Find the energy consumed and the cost of use. 5

20. A bulb is rated 60 W at 220 V. Find the current drawn, resistance of the filament and the energy consumed in 5 hours in kWh and in joules. 5

OR

A circuit has 4 ohm in series with a parallel combination of 6 ohm and 3 ohm across a 12 V battery. Find the total resistance, the current and the voltage across the parallel combination. 5

END OF QUESTION PAPER — Total: 40 Marks

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